

Chapter 3: Perfect-Information Games (aka Game Trees)

ECON 5001

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Based on: *Game Theory* by Giacomo Bonanno

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Everything So Far Happened at Once

In Chapter 2, players chose **simultaneously**, seeing nothing beforehand.

But most interesting situations aren't like that:

- chess
- an offer and a counteroffer
- a firm entering a market and the incumbent responding
- calling a play after seeing the defense line up

Now: players move **in turn**, and everyone sees the whole history.

Moblab: “Centipede Game”

- MobLab time!
- Two players building a firm over time
- As long as players invest, firm doubles in value each day
- Each “turn” one player chooses:
 - **Invest:** keep the firm going.
 - **Cash Out:** sell the firm, take 80% of its value. Game over.
 - Note: If the other person cashes out, you only get 20%!!
- Alternating moves (P1, P2, P1, P2...)
- If value gets to 80, P1 must Cash Out, giving (64, 14)

Feel free to talk! And play it the way you'd actually play it.

How Far Did You Get?

Most frequent # of Turns: _____

Groups that Cashed Out in Turn 1: _____

Groups that made it to the end: _____

Trees

Perfect Information

In a **dynamic game** (or **extensive form**) players move in sequence.

This chapter covers the case of **perfect information**:

whenever it is a player's turn to move,
she knows every move made so far

- **Perfect Information Games:**

- Chess, checkers, Connect Four, bargaining, sequential legislative agenda/votes

- **Imperfect Information Games:**

- Stratego, Battleship, Mastermind, patent races, dynamic attacker/defender games
- Any game with a hidden move by “Nature” (see Chapter 4)
 - Poker, auctions, bidding wars, diplomatic negotiations w/ private info

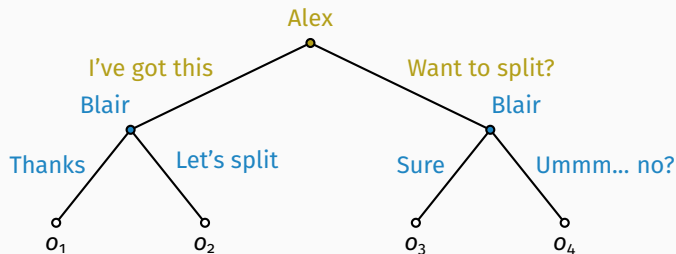
Rooted Directed Trees

We represent these games using **rooted directed trees**.

A **rooted directed tree** is a set of **nodes** joined by directed **edges**.

“Directed” edge: *from* one node *to* another

Ex: Alex and Blair are out to dinner. The check arrives.



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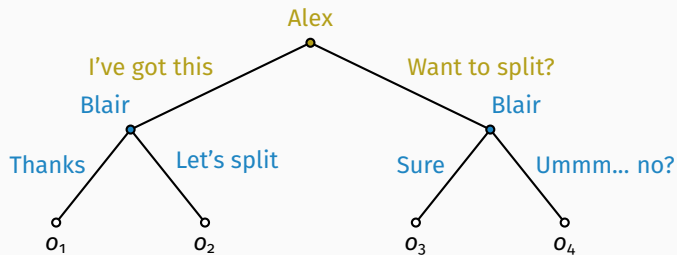
- The **root** node has no edge leading into it
- Every other node has **exactly one** edge leading into it
 - *Branches out*, like a tree
 - Contains *no cycles/loops*
- **Terminal nodes** have no edges going out
 - They are the various game endings
 - In a game, each has an outcome o attached to it
- there is a **unique path** from the root to any other node

Set of **terminal nodes**: Z . All others: “**Decision nodes**” D . Then define the set of **all nodes**:

$$X = D \cup Z$$

Each node $x \in X$ represents a “history” of the game so far

Back to the Example



$D =$

$Z =$

$X =$

Definition: Extensive Form With Perfect Information

A **finite extensive form with perfect information** consists of:

- A finite rooted directed tree
- A set of players $I = \{1, \dots, n\}$
 - Each decision node is assigned to one player
- A set of actions A
 - Each edge is assigned one action
 - Two edges from the same node $\Rightarrow$ different actions
- A set of outcomes O
 - Each terminal node is assigned one outcome

Note: no **preferences** yet! This is a frame, not a game. Outcomes only.

Dissolving the Partnership

Nocterra Brewing Company shut down.

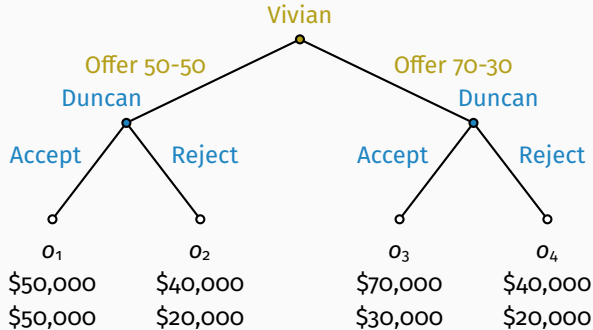
Owners: Vivian and Duncan

Suppose remaining assets are valued at \$100,000. How to split it?

- The charter says the senior partner, **Vivian**, proposes a split
- **Duncan** can Accept, or Reject and take it to court
- In court: each side pays \$20,000 in legal fees, and the verdict will 60% to Vivian, 40% to Duncan (no uncertainty)
- Vivian is allowed only two possible offers: **50-50** or **70-30**

So rejecting yields Vivian $\$60,000 - \$20,000 = \$40,000$
and Duncan $\$40,000 - \$20,000 = \$20,000$.

The Frame



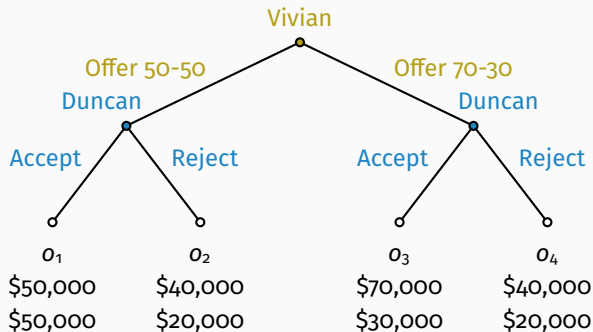
Top number is Vivian's (P_1), bottom number is Duncan's (P_2).

For now let's assume *selfish & greedy* preferences.

Backward-Induction Reasoning

Let's build up a new solution concept

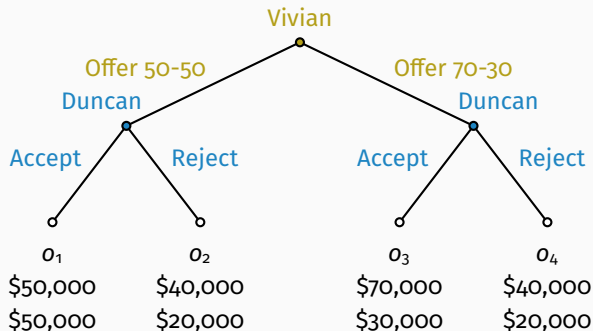
Backwards induction: Start at the *end* and work back toward the root.



Backward-Induction Reasoning

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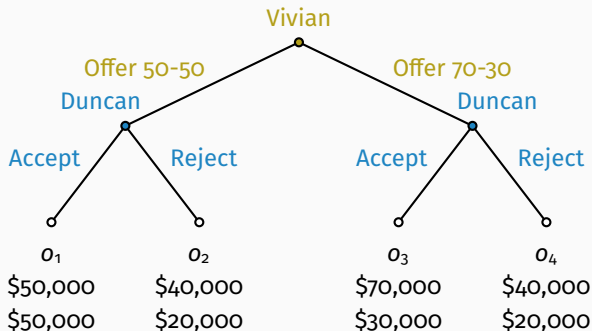


- Duncan accepts at both nodes

Backward-Induction Reasoning

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Backwards induction: Start at the *end* and work back toward the root.

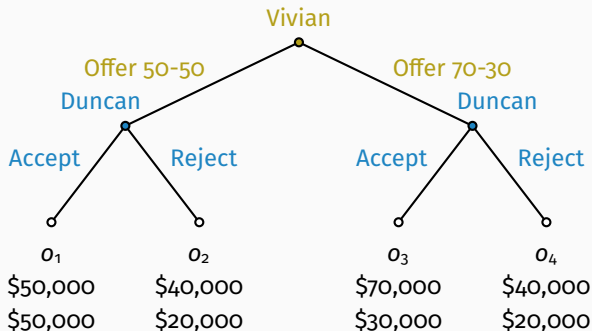


- Duncan accepts at both nodes
- Knowing this, Vivian offers 70-30

Backward-Induction Reasoning

Let's build up a new solution concept

Backwards induction: Start at the *end* and work back toward the root.



- Duncan accepts at both nodes
- Knowing this, Vivian offers 70-30

Another solution concept built on “common knowledge of rationality”

Non-Selfish Preferences?

What if they are *not* selfish & greedy??

Suppose instead Duncan thinks they worked just as hard as Vivian, and that the only fair split is 50-50. They may feel strongly enough that they would give up \$10,000 to refuse an unfair offer.

That is: $o_4 \succ_{\text{Duncan}} o_3$.

This will change the predicted solution:

Duncan Accepts 50-50 but *Rejects* 70-30. So now Vivian offers 50-50.

A tree with *outcomes* at the terminal nodes is a **frame**.

Replace them with the players' *utility values* and you have a **game**.

Adding Preferences: An Example

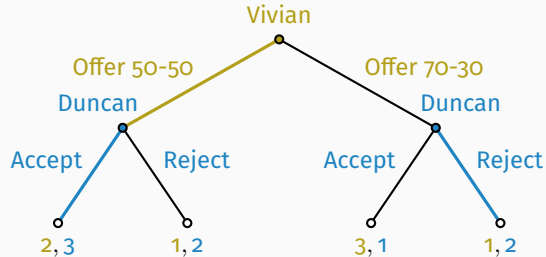
Suppose Vivian is selfish and greedy: $O_3 \succ_1 O_1 \succ_1 O_2 \sim_1 O_4$

Duncan cares about fairness: $O_1 \succ_2 O_2 \sim_2 O_4 \succ_2 O_3$

	O_1	O_2	O_3	O_4
U_1 (Vivian)	2	1	3	1
U_2 (Duncan)	3	2	1	2

Now replace each outcome with its payoff pair to get the actual **game**

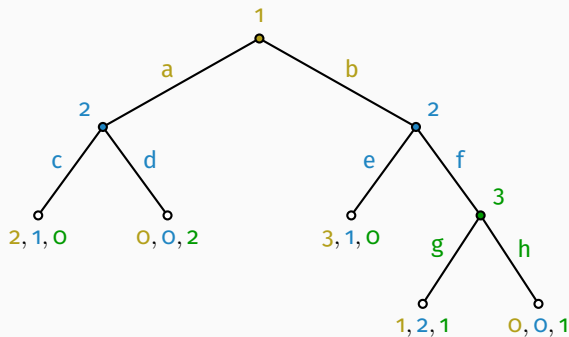
The Game



Colored edges show the backward-induction action choices.
Vivian now anticipates 70-30 will be rejected, offers 50-50.

Backward Induction: How to Handle Ties

With ties, there can be multiple backwards induction predictions:

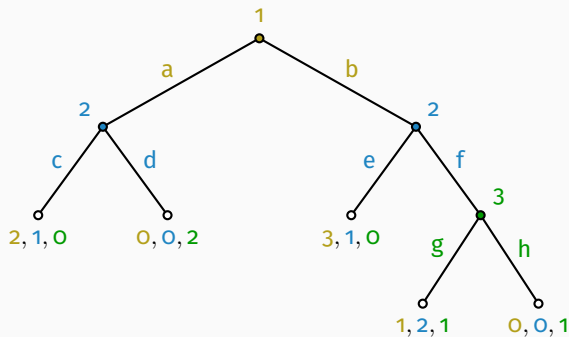


Payoffs in order: Player 1, Player 2, Player 3.

Strategies

Actions vs. Strategies

A **very** important distinction: actions vs. strategies



Actions:

$A_1 =$

$A_2 =$

$A_3 =$

Strategies: “complete plans of actions”

$S_1 =$

$S_2 =$

$S_3 =$

A Strategy Is a Complete Plan

Definition

A **strategy** for a player in a perfect-information game is a list of *actions*, **one for each decision node of that player**.

Not “what I will do” but “what I *would* do everywhere I might move.”

In the previous game, Player 2 has two decision nodes, so a strategy is a pair. Before the game starts she doesn't know whether Player 1 will pick *a* or *b*, so she must plan for both:

(c, e) (c, f) (d, e) (d, f)

Four strategies, not two.

Counting Strategies

Suppose Player 1 has three decision nodes in some game:

- at the first, three choices a_1, a_2, a_3
- at the second, two choices b_1, b_2
- at the third, four choices c_1, c_2, c_3, c_4

A strategy fills in three blanks:

$$\left(\underbrace{\hspace{2cm}}, \underbrace{\hspace{2cm}}, \underbrace{\hspace{2cm}} \right)$$

one of a_1, a_2, a_3 one of b_1, b_2 one of $c_1, \dots, c_4$

So her strategy set is a Cartesian product of the action sets she faces:

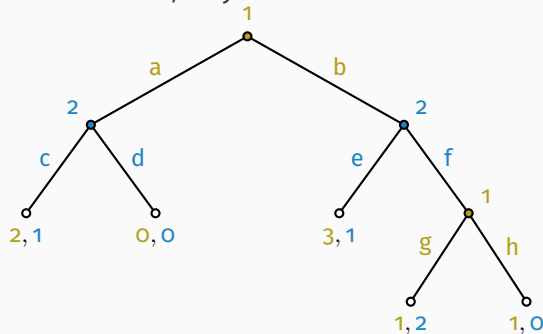
$$S_1 = \{a_1, a_2, a_3\} \times \{b_1, b_2\} \times \{c_1, c_2, c_3, c_4\}$$

$$3 \times 2 \times 4 = 24 \text{ strategies}$$

Strategy sets get large fast! Because they're a complete plan.

Don't Throw Out Impossible Strategies

Here, Player 1 moves twice:



There are 2 backwards-induction predictions:

But strategy (a, g) is impossible! Playing a makes g irrelevant.

But let's keep it. Player 2's choice of f depended on g . Backwards induction requires a specification of how you play *everywhere*.

Other Justifications

Other ways to justify specifying (a, g) (not just (a)):

1. **Mistakes.** She's cautious enough to plan for the possibility that she intends a but ends up playing b .
2. **Delegation.** A strategy is a set of instructions handed to someone playing on her behalf, who needs to know what to do everywhere.
3. **Beliefs.** A strategy isn't really Player 1's plan at all, but Player 2's *belief* about what Player 1 would do.

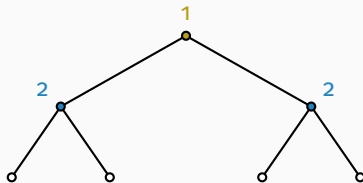
But the main reason: **Backwards induction needs it**

On Path and Off Path

A strategy profile traces one path from the root to a terminal node.

- Nodes on that path are **on the path of play** (“on path”)
- Every other node is **off the path of play** (“off path”)

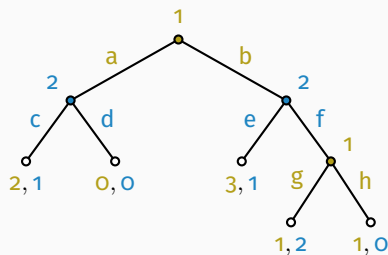
A strategy must still say what to do off path. Even if certain actions aren't on-path, they make the on-path actions rational.



From Tree to Table

Every strategy profile determines a unique path, hence a unique terminal node, hence a unique payoff vector.

So every perfect-information game has an **associated strategic form**.



		Player 2			
		ce	cf	de	df
Player 1	ag	2, 1	2, 1	0, 0	0, 0
	ah	2, 1	2, 1	0, 0	0, 0
	bg	3, 1	1, 2	3, 1	1, 2
	bh	3, 1	1, 0	3, 1	1, 0

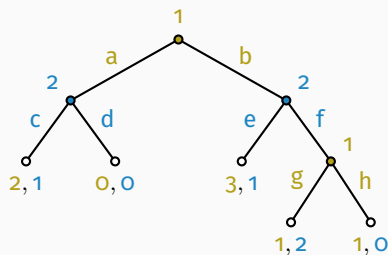
Note: It's okay to write strategies as *ag* and *ah* instead of (a, g) and (a, h)

The top two rows are identical because *ag* and *ah* are *redundant*

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	ah	2, <u>1</u>	<u>2</u> , 1	0, 0	0, 0
	bg	<u>3</u> , 1	1, <u>2</u>	<u>3</u> , 1	<u>1</u> , <u>2</u>
	bh	<u>3</u> , 1	1, 0	<u>3</u> , 1	<u>1</u> , 0

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Five Equilibria, Two Solutions

That game has **five** Nash equilibria:

(ag, cf) (ah, cf) (bg, df) (bh, ce) (bh, de)

It has **two** backward-induction solutions:

- (ag, cf)
- (bh, ce)

Both are Nash equilibria.

The other three are not backward-induction solutions.

That gap is the whole point of the next section.

Solution vs. Outcome

A backward-induction **solution** is a *strategy profile*. It describes a complete plan of actions, including at nodes that never get reached (“off path” nodes).

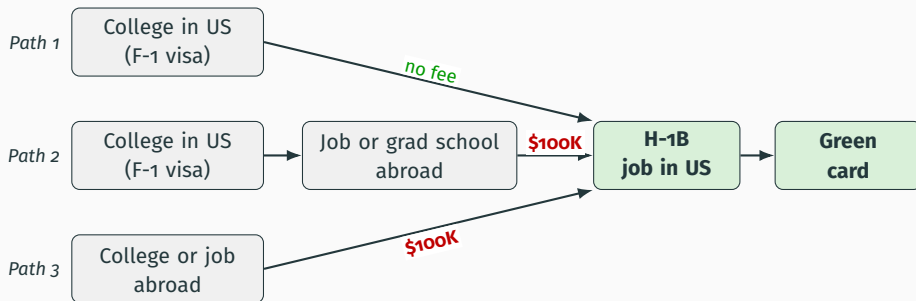
A backward-induction **outcome** is the *sequence of moves actually made*.

Solution (ag, cf) outcome ac , payoffs $(2, 1)$

Solution (bh, ce) outcome be , payoffs $(3, 1)$

Real-World Example: Skilled Immigration to the US

Three typical paths to a green card for skilled workers. All three merge at the H-1B.



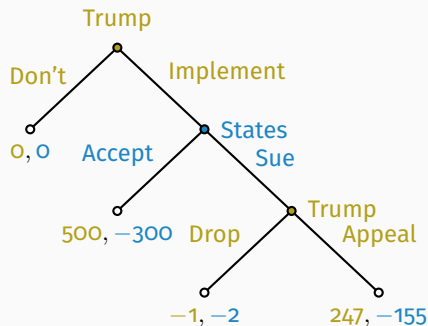
Sept. 2025: an executive order charges employers a **\$100,000 fee** for a new H-1B worker who is *outside* the US. A student switching from F-1 to H-1B *inside* the US (Path 1) is exempt.

The Lawsuit

- **Dec. 2025:** 20 states, led by California and Massachusetts, sue in U.S. District Court in Boston. (Ohio is not one of them.)
- Their argument: the fee is really a **tax**. Taxes are set by **Congress**, not the President.
- **June 2026:** Judge Leo Sorokin agrees and strikes down the fee: it “imposes a tax on H-1B petitions without the requisite delegation by Congress.”
- **July 24, 2026:** The First Circuit Court of Appeals refuses to pause that ruling while the government appeals, claiming the gov’t is unlikely to win
- **Sept. 2026:** The appeal continues in the First Circuit, and will likely go to the Supreme Court.

To keep things simple, we’ll fold the First Circuit Court and Supreme Court into one “appeal” move, and assume the District and Circuit Courts side with the States, while the Supreme Court is a 50-50 coin flip.

The Game



Payoffs in \$ millions (rough guesses):

- Fee stands: Trump +500, States -300
- Fee struck down: both get 0
- Legal fees if States sue: Trump -1, States -2
- Extra legal fees if Trump appeals: Trump -2, States -3

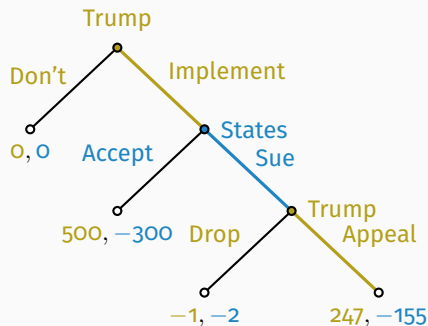
After an appeal, the Supreme Court is a coin flip, so we average:

- Trump: $\frac{1}{2}(500) + \frac{1}{2}(0) - 1 - 2 = 247$
- States: $\frac{1}{2}(-300) + \frac{1}{2}(0) - 2 - 3 = -155$

Drop: fee struck down, but both paid for the district court case.

What does backward induction predict?

The Game



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Backward induction: **Implement, Sue, Appeal.**

What If the States Expect to Lose?

Let p be the chance the Supreme Court sides with the States.

States' average payoff:

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States' average payoff:

- Sue (and Trump appeals): $(0)p + (-300)(1 - p) - 5$
- Accept: -300

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- Accept: -300
- Suing is (weakly) better when

$$(0)p + (-300)(1 - p) - 5 \geq -300$$

$$-300 + 300p - 5 \geq -300$$

$$300p - 5 \geq 0$$

$$300p \geq 5$$

$$p \geq 1.\overline{666}\%$$

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Legal fees are tiny next to the stakes. Even a long shot is worth suing over.

What If the State Expect to Lose?

The model also leaves out reasons that make suing even more attractive:

- **Delay.** The fee is blocked while the case goes on.
- **Politics.** Attorneys general gain from being *seen* fighting.

Backward Induction vs. Nash Equilibrium

The Theorem

Theorem. Every backward-induction solution of a perfect-information game is a Nash equilibrium of the associated strategic form.

Suppose player i deviates from B.I. at some node:

- If that deviation is **on path** then B.I. already checks that it gives a lower payoff.
- If that deviation is **off path** then it's irrelevant, so not strictly better or worse.
- Thus: no deviation is **strictly** better

So backward induction is a **refinement** of Nash equilibrium:

$$BI(G) \subseteq NE(G) \quad \forall G$$

Which Nash Equilibria are Not B.I.?

Which Nash equilibria are ruled out by B.I.?

Usually the ones resting on an **incredible threat**.

The Entry Game

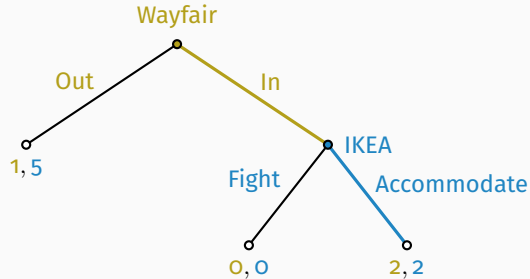
IKEA has been the only big furniture store at Polaris for years, earning \$5 million/yr
Wayfair had to decide whether to open across I-71.

- If Wayfair stays **out**, it earns \$1 million elsewhere
- If Wayfair enters, IKEA can **fight**, a price war leaving both at zero
- Or IKEA can **accommodate** and share, and both make \$2 million

Both firms are selfish and greedy.

Suppose IKEA announces: “if Wayfair opens in Polaris we will bury them on price.”
Should Wayfair believe it?

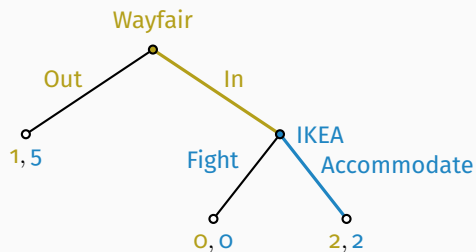
The Entry Game



First number Wayfair, second number IKEA.

Backward induction: faced with entry, IKEA compares 0 to 2. So it accommodates. Knowing that, Wayfair enters.

But There Are Two Nash Equilibria



		IKEA	
		Fight	Accom.
Wayfair	In	0, 0	<u>2</u> , <u>2</u>
	Out	<u>1</u> , <u>5</u>	1, <u>5</u>

- (In, Accommodate), the backward-induction solution
- (Out, Fight), also a Nash equilibrium!

It is an **incredible threat**: if entry actually happened, IKEA would not want to carry it out.

Lesson: Off-path play can be very important!

Backwards Induction: Off-path play **must** be credible

Nash Equilibrium: Off-path play can be non-credible

Selten's Chain-Store Game

IKEA also has a store in Cincinnati, and Wayfair could expand there next. So the entry decision happens **one city at a time**, with everyone watching what happened in the earlier one.

IKEA makes the argument in CBus:

Fighting you costs me \$2 million today. But Cincy is watching. If they see me fight, they may stay out, and then I make $\$(0+5) = \5 million across the two cities. If I accommodate you here, they enter there too, and I make $\$(2+2) = \4 million. So it is in my interest to fight.

Sounds right. But let's do backwards induction across cities...

The Reply

Backwards induction: start with Cincinnati

- Whatever happens in CBus, Cincy is the **last** interaction
- Nobody is watching after that, so no reputation is left to build
- So Cincy is just the one-shot game, and IKEA accommodates
- Thus (according to B.I.), Cincy goes the same way *regardless* of CBus
- So fighting in CBus buys IKEA **nothing**:
 $\$(0 + 2) = \2 million against $\$(2 + 2) = \4 million

Entry happens in both cities. IKEA accommodates both.

Backward induction means you can't build a reputation for being "tough"

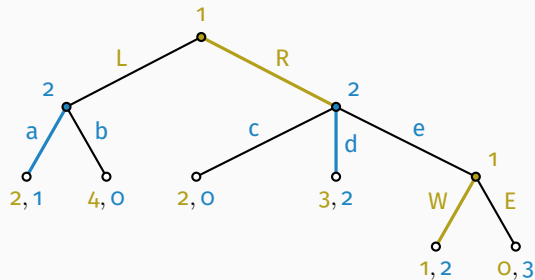
How can we build a reputation story?

- The reputation story needs **uncertainty**: the Cincy entrant must be unsure what kind of opponent IKEA is
- In a perfect-information game there is no uncertainty, so there is no reputation-building
 - Uncertainty about your opponent's "type": Chapter 15

We'll come back to this with better tools. For now, note the limitation.

We Know that $SNE(G) \subseteq BI(G) \subseteq NE(G) \subseteq IDSDS(G)$
But how does $BI(G)$ relate to $IDWDS(G)$?

A Backward-Induction Solution That IDWDS Deletes

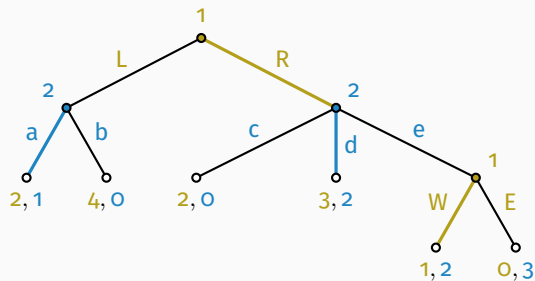


After **R**, Player 2 is **indifferent** between **d** and **e** (both pay 2, given **W**). So there are two BI solutions:

- $((R, W), (a, d))$, payoffs (3, 2) (shown)
- $((L, W), (a, e))$, payoffs (2, 1)

Bonanno, Exercise 3.8.

A Backward-Induction Solution That IDWDS Deletes



After **R**, Player 2 is **indifferent** between **d** and **e** (both pay 2, given **W**). So there are two BI solutions:

- $((R, W), (a, d))$, payoffs (3, 2) (shown)
- $((L, W), (a, e))$, payoffs (2, 1)

But **ae** **weakly dominates** **ad**: equal against **L** and **RW**, strictly better against **RE** (3 vs. 2).

e is insurance in case Player 1 slips and plays **E**. IDWDS deletes **ad**, and the shown solution goes with it.

Bonanno, Exercise 3.8.

The Same Game in Strategic Form

(*LW* and *LE* are identical rows, merged here into *L*.)

		Player 2					
		ac	ad	ae	bc	bd	be
Player 1	L	2, 1	2, 1	2, 1	4, 0	4, 0	4, 0
	RW	2, 0	3, 2	1, 2	2, 0	3, 2	1, 2
	RE	2, 0	3, 2	0, 3	2, 0	3, 2	0, 3

Shaded: the two backward-induction solutions, (*L*, *ae*) and (*RW*, *ad*).

The Same Game in Strategic Form

(*LW* and *LE* are identical rows, merged here into *L*.)

		Player 2					
		ac	ad	ae	bc	bd	be
Player 1	L	2, 1	2, 1	2, 1	4, 0	4, 0	4, 0
	RW	2, 0	3, 2	1, 2	2, 0	3, 2	1, 2
	RE	2, 0	3, 2	0, 3	2, 0	3, 2	0, 3

Round 1: *RE* is weakly dominated by *RW*; *ae* weakly dominates every other column of Player 2. All are deleted at once.

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(*LW* and *LE* are identical rows, merged here into *L*.)

		Player 2					
		ac	ad	ae	bc	bd	be
Player 1	L	2, 1	2, 1	2, 1	4, 0	4, 0	4, 0
	RW	2, 0	3, 2	1, 2	2, 0	3, 2	1, 2
	RE	2, 0	3, 2	0, 3	2, 0	3, 2	0, 3

Round 2: against *ae*, *L* (2) beats *RW* (1). IDWDS leaves only (*L*, *ae*): **two** BI solutions, but IDWDS keeps just **one**.

Relationship With Other Solution Concepts

What we knew:

- $SNE(G) \subseteq NE(G) \subseteq IDSDS(G)$
- $SNE(G) \subseteq IDWDS(G) \subseteq IDSDS(G)$

Where does Backward Induction ($BI(G)$) land?

1. $SNE(G) \subseteq BI(G)$ (optional: prove this)
2. $BI(G) \subseteq NE(G)$
3. $BI(G) \cap IDWDS(G) \neq \emptyset$ (optional: prove this)

Thus:

- $SNE(G) \subseteq [BI(G) \cap IDWDS(G)] \subseteq BI(G) \subseteq NE(G) \subseteq IDSDS(G)$

Win, Lose, or Draw

Win-Lose Games

Two players, two outcomes: Player 1 wins (W_1) or Player 2 wins (W_2). **No draws.**

Player 1 strictly prefers W_1 ; Player 2 strictly prefers W_2 . Use payoffs $(1, 0)$ and $(0, 1)$.

Theorem. In every finite two-player win-lose game with perfect information, **one of the two players has a winning strategy.**

A winning strategy guarantees a win *no matter what the opponent does.*

Note what it doesn't say: *which* player, or what the strategy is.

Let's Play

Two players take turns choosing a number from $\{1, 2, \dots, 10\}$.

Running total starts at 0. **First to reach 100 or more wins.**

Player 1 moves first.

How to play?

Why I Was Always Going to Lose That

The tree is far too big to draw: 10,000 nodes just for the first four moves. But we can reason backward anyway.

Call a total a **losing position** if whoever has to move from it cannot win.

- **89** is a losing position: from 89 you can reach only 90, ..., 99, and from any of those I can hit exactly 100
- **78** is the next one down: from 78 you must reach 79, ..., 88, and from any of those I can get to 89
- Keep going: 89, 78, 67, 56, 45, 34, 23, 12, **1**

Player 1's Winning Strategy

Start with **1**.

Then every turn, choose **$11 - n$** , where n is what your opponent just chose.

Each pair of moves adds exactly 11, so the total marches 1, 12, 23, $\dots$, 89, 100.

Notice this is a genuine *strategy*: it says what to do at every position Player 1 could face, not just the ones on the winning path.

Adding a Draw

Now three outcomes: W_1 , W_2 , and a draw D , with

$$W_1 \succ_1 D \succ_1 W_2 \quad \text{and} \quad W_2 \succ_2 D \succ_2 W_1$$

Use utilities $(1, 0, -1)$

Theorem. Every such game falls into exactly one of three categories:

1. Player 1 has a strategy guaranteeing W_1
2. Player 2 has a strategy guaranteeing W_2
3. Player 1 can guarantee *at least* a draw, and so can Player 2, so if both play well it's a draw

The proof is the same backward-induction argument, with three possible markings instead of two.

Which Category?

- **Tic-Tac-Toe:** category 3. Solved: perfect play draws.
- **Checkers:** category 3. Solved in 2007, after 18 years of work.
- **Chess:** _____

The theorem says chess is in *one* of the three categories. It does not say which, and nobody knows.

Existence proofs are like that.

Back to the Centipede

Apply backward induction to the game you played:

- At the **last** decision, taking beats passing. So take.
- At the second-to-last, you know the other player will take next, so passing gets you the smaller share. So take.
- Keep unraveling ... all the way to the first move.

Prediction: the very first player takes immediately.

Our numbers said: _____

So Who's Wrong?

Three honest possibilities:

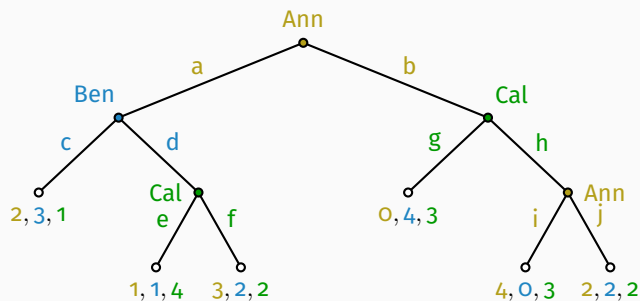
- The **payoffs** are wrong. Players may care about the other person or about fairness, so the game isn't the one we wrote down
- The **reasoning** is wrong. Players may not carry the unraveling through that many steps
- The **beliefs** are wrong. I might reason perfectly and still pass if I doubt that *you* will

All three are live. We'll get the machinery to separate them later.

Let's Work One Out

Planning Spring Break

Ann, Ben, and Cal share a house and settle the trip in turn, each seeing every decision made before theirs.



Payoffs in order: Ann, Ben, Cal.

Planning Spring Break

1. How many strategies does each player have?
2. Run backward induction. Where does a player face a tie?
3. Write out **both** backward-induction solutions, as strategy profiles.
4. What are the two backward-induction outcomes?

What Have We Learned, Charlie Brown?

- Perfect-information games are drawn as **rooted trees**; a frame becomes a game once preferences are added
- **Backward induction** works from the leaves inward; ties give several solutions
- A **strategy** covers all your decision nodes, even unreachable ones
- Every backward-induction solution is a **Nash equilibrium**; the extra Nash equilibria typically rest on **incredible threats**
- In finite two-player win-lose games, **somebody** has a winning strategy

Next: what happens when players *don't* see the whole history.

Reading & Practice

- Bonanno, *Game Theory*, Chapter 3
- Exercises: §3.7.1 (trees and frames), §3.7.2 (backward induction), §3.7.3 (strategies), §3.7.4 (two-player games)
- Exercise 3.8 shows IDWDS missing a backward-induction solution
- Solutions in §3.8